# Quiz 1 - DSC 10, Fall 2025

Full Name:

PID:

Quiz Time: ◯ 2PM ◯ 3PM ◯ 4PM ◯ 5PM

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**Instructions:**

- This quiz consists of 5 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.
- Assume we have already run `import babypandas as bpd` and `import numpy as np`.

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By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

# Version A

Please do not open your quiz until instructed to do so.

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You are the newest employee at Nintendo and you are tasked with creating the next Super Smash Brothers video game! Games in this famous series involve battles between characters from many different franchises, like Pokemon, Mario, and Legend of Zelda. You've been given the roster of characters (known as fighters) in the current game, but unfortunately, due to budget cuts, you'll need to eliminate some of them.

The `smash` DataFrame, whose first few rows are shown below, includes information on all fighters in the current game. For each fighter, you have their `"Name"` (string), `"Species"` (string), `"Franchise"` of origin (string), `"Year"` of origin (int), and primary `"Weapon"` (string).

| | Name | Species | Franchise | Year | Weapon |
|---|---|---|---|---|---|
| 0 | Mario | Human | Mario | 1981 | Hands/Feet |
| 1 | Sephiroth | Human | Final Fantasy | 1997 | Sword |
| 2 | Steve | Human | Minecraft | 2011 | Weapon (Other) |
| 3 | Kirby | Creature | Kirby | 1992 | Hands/Feet |
| 4 | Donkey Kong | Animal | Donkey Kong | 1981 | Hands/Feet |

## Question 1

Nintendo management sends you some information about your new job, but there seems to be an encryption puzzle blocking you from opening the file. To undo the encryption and open the file, **determine the data type** of each expression below, or indicate that the expression produces an error.

**a)** `smash.shape[0] / 1`

◯ int ◯ float ◯ str ◯ bool ◯ DataFrame ◯ error

**b)** `str(len(smash)) * 2.0`

◯ int ◯ float ◯ str ◯ bool ◯ DataFrame ◯ error

**c)** `smash.get("Year").iloc[0] == "1981"`

◯ int ◯ float ◯ str ◯ bool ◯ DataFrame ◯ error

**d)** `smash.take([0])`

◯ int ◯ float ◯ str ◯ bool ◯ DataFrame ◯ error

## Question 2

For diversity, we need to cut fighters from the most common `"Species"`. Find the `"Species"` with the most fighters (i.e. the `"Species"` most represented in `smash`).

```
smash.groupby("Species").___(a)___().sort_values(by="Year").___(b)___
```

(a): _____ (b): _____

## Question 3

You decide to eliminate some of the fighters that have `"Sword"` as their `"Weapon"`. **Select all** expressions that evaluate to the total number of such fighters.

☐ `smash[smash.get("Weapon") == "Sword"].shape[0]`

☐ `smash.get("Weapon").count("Sword")`

☐ `(smash.get("Weapon") == "Sword").shape[0]`

☐ `smash[smash.get("Weapon") == "Sword"].sum()`

☐ `smash.groupby("Weapon").count().get("Franchise").loc["Sword"]`

☐ None of the above.

## Question 4

Alongside the unveiling of your new game, you'll want to announce a new fighter to build hype. The code below calculates the name of this new fighter and stores it in the variable `new`. In the box, write in the value of the `new` variable after the code executes. Write clearly!

```
lu = "luigi"
dk = "dk"
new = "crash"
dk.replace("k", "kong")
new = ((lu + "t") * len(dk)).title()
new = new
```

`new:`

## Question 5

For the opening trailer, your boss needs you to find out two facts about Super Smash Brothers history. Write one line of Python code that evaluates to each quantity below.

**a)** The median origin `"Year"` of all fighters in `smash`.

**b)** The name of the fighter with the earliest origin `"Year"` (or in the case of a tie, any fighter from the earliest origin `"Year"`).

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**Reminders:**

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.